Exact Autonomy Dimension of Tetrahedral Windmills

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Abstract

Let W_r be the graph formed from r copies of K_4 sharing a common vertex. Its autonomy dimension is the minimum number of binary observations that separate all vertices and individually project simple random walk to a Markov chain for every starting state. An earlier note asked for the exact value, having established only a first-order asymptotic formula. We answer that question for every r. The answer is the inverse of an explicit binomial-sum capacity formula, with three cases according to the coordinate count modulo three. The proof converts autonomous codes into packings of three-block partitions, bounds these packings by weights on subset sizes, and attains the bounds using a classical almost-regular hypergraph decomposition theorem. We also obtain the second-order formula

$$\operatorname{adim}(W_r) = \frac{\log_2(3r) + \frac{1}{2}\log_2\log_2(3r)}{H_2(1/3)} + O(1).$$

Integer certificates and explicitly constructed codes accompany the proof.

Keywords. Strong lumpability; autonomous coordinates; windmill graph; set partitions; hypergraph decomposition.

1 The question and its exact answer

Let G be a finite connected simple graph with at least two vertices. A nonconstant function $f : V(G) \to \{0, 1\}$ is *autonomous* if the fraction of neighbors of opposite color is a constant a on $f^{-1}(0)$ and a constant b on $f^{-1}(1)$. Its observed random walk has transition matrix $\left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$ for every starting state. The *autonomy dimension* $\operatorname{adim}(G)$ is the minimum number of autonomous binary functions whose joint values distinguish all vertices; it is infinity if no such family exists.

Write W_r for r copies of K_4 with one vertex identified to a common hub o. There are $3r + 1$ vertices. Removing the hub leaves r triangles. The October 7 research note [1], Section 7, asked for the exact autonomy dimension of this family. This is the specific open question addressed here; it is a question from that research program, not a claim to settle an independently established named conjecture.

For a fixed integer $d \geq 2$, set

$$B_j = \binom{d}{j}, \qquad S_j = \sum_{i=0}^{j} \binom{d}{i},$$

where $B_j = 0$ and $S_j = 0$ when $j < 0$. Their dependence on d is implicit.

Theorem 1.1 (Exact capacity and dimension)**.** *Let M_d be the largest r for which W_r has a separating autonomous code using at most d coordinates. Then*

$$M_d = \begin{cases} S_{k-2} + \lfloor (2B_{k-1} + B_k)/3 \rfloor, & d = 3k, \quad k \geq 1, \\ S_{k-1} + \lfloor B_k/2 \rfloor, & d = 3k+1, \quad k \geq 1, \\ S_{k-2} + B_k + \lfloor 3B_{k-1}/4 \rfloor, & d = 3k+2, \quad k \geq 0. \end{cases} \tag{1.1}$$

Consequently, for every integer $r \geq 1$,

$$\boxed{\operatorname{adim}(W_r) = \min\{d \geq 2 : r \leq M_d\}.} \tag{1.2}$$

d	2	3	4	5	6	7	8	9	10	11	12	13
M_d	1	1	3	5	10	18	35	62	116	218	390	735

Table 1: Exact capacities. For example, $\operatorname{adim}(W_{18}) = 7$ and $\operatorname{adim}(W_{19}) = 8$. The repeated capacity at $d = 3$ means that no windmill has autonomy dimension three.

Formula (1.2) is an explicit evaluation rule involving only binomial coefficients and integer arithmetic. It requires no search over graph colorings. The proof below relies on Bryant's established hypergraph balancing theorem [2]; the decomposition theorem itself is not new. Historical priority of the exact partition-packing formula or the graph-theoretic invariant has not been established by a comprehensive literature review.

2 From graph coordinates to partitions

We first include the local classification needed for the reduction.

Lemma 2.1. *Normalize a nonconstant autonomous function on W_r by $f(o) = 0$. Then $f^{-1}(1)$ contains exactly q vertices from each outer triangle, for one common $q \in \{1, 2, 3\}$. Conversely every such function is autonomous, with $a = q/3$ and $b = (4 - q)/3$.*

Proof. The hub has positive flip probability. Thus no triangle can have all three values zero, since an outer zero vertex in that triangle would have zero flip probability. If triangle i has q_i ones, each of its one vertices has $(4 - q_i)/3$ as its flip probability. Constancy on the one class forces all q_i to agree. Conversely, for a common q, both the hub and every outer zero have flip probability $q/3$, while every one has flip probability $(4 - q)/3$. □

A *three-block packing on* $[d]$ is a family of triples $\{A_i, B_i, C_i\}$ such that each triple consists of pairwise disjoint subsets with union $[d]$, and all subsets appearing anywhere in the family are distinct. One block may be empty, but the empty set can occur at most once in the entire family.

Proposition 2.2 (Packing equivalence)**.** *For $d \geq 2$, W_r has a separating autonomous code using at most d coordinates if and only if there is a three-block packing of size r on $[d]$.*

Proof. Start with a separating code and complement coordinates so that the hub has the all-zero word. By Lemma 2.1, each column selects one, two, or three vertices in every triangle. Delete the columns selecting three; they are constant on all outer vertices, so deleting them preserves separation among outer vertices. In each column selecting two, flip its values on the outer vertices

only. This is a common translation of the outer words and also preserves their distinctness. Every remaining column now has exactly one one in every triangle.

The supports of the three words of a triangle therefore partition the remaining coordinate set, and all $3r$ supports are distinct. If fewer than d coordinates remain, add the missing coordinates to one chosen block in each triple. Restriction to the old coordinates proves that the blocks remain distinct. This yields a packing on $[d]$.

Conversely, a packing uses $3r$ distinct subsets of $[d]$. Since $3r \le 2^d$ and 3 does not divide 2^d, at least one subset T is unused. Assign the zero word to the hub and the incidence words of $A_i \triangle T$, $B_i \triangle T$, $C_i \triangle T$ to triangle i, where $\triangle$ denotes symmetric difference. Every word is nonzero and the words are distinct. For a coordinate outside T, exactly one vertex per triangle has value one; for a coordinate in T, exactly two do. Each coordinate is autonomous by Lemma 2.1. □

The unused subset is essential: simply assigning incidence words and giving the hub zero would fail whenever a block is empty.

3 Upper bounds by weights on subset sizes

Lemma 3.1 (A weight bound)**.** *Suppose $w_0, \dots, w_d \ge 0$ satisfy $w_a + w_b + w_c \ge 1$ whenever $a + b + c = d$ with $a, b, c \ge 0$. Every three-block packing on $[d]$ then has size at most*

$$\left\lfloor \sum_{j=0}^{d} B_j w_j \right\rfloor .$$

For any real $t > d/3$, a valid choice is

$$w_j = \min\left\{1, \max\left\{0, \frac{t-j}{3t-d}\right\}\right\}.$$

Proof. Each triple has total weight at least one, and no subset is reused. Summing over its blocks proves the first assertion. For the displayed weights, if any one of the three untruncated values is at least one, the corresponding weight is one. Otherwise truncating negative values to zero can only increase their sum, whose untruncated value is $(3t - a - b - c)/(3t - d) = 1$. □

For $d = 3k$, take $t = k + 1$. The nonzero weights are

$$w_j = 1 \ (j \le k-2), \qquad w_{k-1} = 2/3, \qquad w_k = 1/3.$$

For $d = 3k + 1$, the same choice gives

$$w_j = 1 \ (j \le k-1), \qquad w_k = 1/2.$$

For $d = 3k + 2$, take $t = k + 2$, giving

$$w_j = 1 \ (j \le k-2), \quad w_{k-1} = 3/4, \quad w_k = 1/2, \quad w_{k+1} = 1/4.$$

Negative indices are omitted. In the last case $B_{k+1} = 2B_k$. Lemma 3.1 therefore gives exactly the three upper bounds in (1.1).

4 Constructions attaining the bounds

We use the following specialization of an established result.

Lemma 4.1 (Prescribed block sizes)**.** *Let (a_i, b_i, c_i) be nonnegative integer triples, each summing to d. They can be realized by a three-block packing on $[d]$ if and only if, for each j, the number of occurrences of j among all the triples is at most $\binom{d}{j}$.*

Proof. Necessity counts distinct subsets of each size. For sufficiency, apply Bryant [2, Corollary 3] to the hypergraph of all subsets of $[d]$, with one prescribed color class for each triple. It supplies distinct subsets with those size counts and makes every color class almost regular. Each such class has total degree d on d vertices, so all its degrees are one. Its three subsets thus form a partition. □

To apply the lemma, it suffices to prescribe block sizes and verify the binomial capacities. We give these lists explicitly. Empty sums below are zero. We repeatedly use

$$\frac{B_{j-1}}{B_j} = \frac{j}{d-j+1}, \qquad B_\ell \geq B_j \quad \text{if } j \leq \ell \leq d-j. \tag{4.1}$$

In the ranges below, the ratios in the lower tail are at most $1/2$. In particular, $S_{k-3} \leq B_{k-2}$ for $k \geq 2$, and $S_{k-2} \leq B_{k-1}$ for $k \geq 1$.

4.1 The case $d = 3k$

For each $0 \leq j \leq k-2$, prescribe B_j triples of type

$$(j, k+1, 2k-1-j).$$

Also prescribe B_{k-1} triples of type $(k-1, k, k+1)$ and

$$z = \left\lfloor \frac{B_k - B_{k-1}}{3} \right\rfloor$$

triples of type (k, k, k). Their number is

$$S_{k-2} + B_{k-1} + z = S_{k-2} + \left\lfloor \frac{2B_{k-1} + B_k}{3} \right\rfloor.$$

The capacities at ranks at most k hold by construction. For $k \geq 2$, the number of blocks of size $k+1$ is

$$L = S_{k-3} + 2B_{k-2} + B_{k-1} \leq 3B_{k-2} + B_{k-1} \leq \tfrac{5}{2} B_{k-1} \leq B_{k+1}.$$

The last inequality follows from $B_{k+1}/B_{k-1} = 2(2k+1)/(k+1) \geq 3$. At any larger rank ℓ, the demand is a single B_j with $j \leq \ell \leq d-j$, so (4.1) applies. When $k = 1$, the list is just one triple of type $(0, 1, 2)$ and the capacities are immediate. Lemma 4.1 gives the packing.

4.2 The case $d = 3k+1$

For each $0 \leq j \leq k-1$, prescribe B_j triples of type

$$(j, k+1, 2k-j),$$

and prescribe $\lfloor B_k/2 \rfloor$ triples of type $(k, k, k+1)$. This gives $S_{k-1} + \lfloor B_k/2 \rfloor$ triples.

Only rank $k+1$ requires a combined capacity check. Its demand is

$$L = S_{k-2} + 2B_{k-1} + \lfloor B_k/2 \rfloor \le 3B_{k-1} + \tfrac{1}{2}B_k < B_{k+1},$$

because

$$\frac{B_{k-1}}{B_k} = \frac{k}{2k+2}, \qquad \frac{B_{k+1}}{B_k} = \frac{2k+1}{k+1}.$$

Every larger rank receives a single demand B_j covered by (4.1). All other capacities are immediate.

4.3 The case $d = 3k+2$

First suppose $k \ge 1$. Put

$$A = B_{k-1}, \qquad B = B_k, \qquad x = \left\lfloor \frac{A+1}{4} \right\rfloor, \qquad y = B - \left\lceil \frac{A}{2} \right\rceil.$$

For $0 \le j \le k-2$, prescribe B_j triples of type $(j, k+2, 2k-j)$. Complete the list with

block-size triple	number of copies
$(k-1, k+1, k+2)$	A
$(k, k, k+2)$	x
$(k, k+1, k+1)$	y.

Here $y \ge 0$, and checking A modulo four gives

$$A + x + y = B + \left\lfloor \frac{3A}{4} \right\rfloor.$$

The number of triples is therefore the last expression in (1.1).

At rank k the demand is $2x + y \le B$; at rank $k+1$ it is $A + 2y \le 2B = B_{k+1}$. For $k \ge 2$, the rank $k+2$ demand satisfies

$$L = S_{k-3} + 2B_{k-2} + A + x \le 3B_{k-2} + \tfrac{3}{2}A \le 3A < B_{k+2}.$$

We used $x \le A/2$, $B_{k-2} \le A/2$, and

$$\frac{B_{k+2}}{A} = \frac{2k+3}{k} \cdot 2 \cdot \frac{2k+1}{k+2} > 4 \quad (k \ge 1).$$

When $k = 1$, the rank $k+2$ demand is just $A + x = 1$, so it also fits. Larger ranks again satisfy (4.1). For $k = 0$ (that is, $d = 2$), one triple of type $(0, 1, 1)$ realizes the claimed capacity one.

All prescribed lists thus meet every binomial capacity. Together with Lemma 4.1, the upper bounds of Section 3 and Proposition 2.2, this completes the proof of Theorem 1.1.

5 A second-order law

The exact formula identifies the term that was hidden inside the $O(\log \log r)$ error in the earlier note.

Corollary 5.1. *Put* $h = H_2(1/3) = \log_2 3 - 2/3$. *Along* $d \equiv s \pmod 3$,

$$M_d \sim c_s \frac{2^{hd}}{\sqrt{d}}, \qquad c_0 = \frac{7}{4\sqrt{\pi}}, \quad c_1 = \frac{9}{4\,2^{1/3}\sqrt{\pi}}, \quad c_2 = \frac{45}{16\,2^{2/3}\sqrt{\pi}}.$$

Consequently, as $r \to \infty$,

$$\boxed{\operatorname{adim}(W_r) = \frac{\log_2(3r) + \frac{1}{2}\log_2\log_2(3r)}{h} + O(1).} \tag{5.1}$$

Proof. Write $d = 3k + s$ with $s \in \{0, 1, 2\}$. For each fixed ℓ, $B_{k-\ell}/B_k \to 2^{-\ell}$. The lower-tail ratios are bounded above by $1/2$, so summing the resulting geometric series gives $S_{k-2}/B_k \to 1/2$ and $S_{k-1}/B_k \to 1$. The three expressions for M_d/B_k therefore tend to $7/6$, $3/2$, and $15/8$, respectively. Stirling's formula yields

$$B_k \sim \frac{3}{2\sqrt{\pi}}\, 2^{-s/3}\, \frac{2^{hd}}{\sqrt{d}},$$

which proves the capacity asymptotics.

Let $d = \operatorname{adim}(W_r)$. For large r, minimality implies $M_{d-1} < r \le M_d$. The three positive constants above are bounded away from zero and infinity, hence $\log_2 r = hd - \frac{1}{2}\log_2 d + O(1)$. This first implies $d \sim (\log_2 r)/h$, and then substituting that equivalence into the logarithmic term proves (5.1). Replacing $\log_2 r$ by $\log_2(3r)$ changes only the bounded term. □

In particular, the logarithmic correction is necessary; it cannot be replaced by a bounded error in the first-order formula.

6 Worked thresholds and reproducible certificates

For seven coordinates, the upper-bound weights are one at subset sizes zero and one, one half at size two, and zero at larger sizes. Thus

$$M_7 \le \left\lfloor 1 + 7 + \tfrac{1}{2}\binom{7}{2} \right\rfloor = 18.$$

The construction uses one triple of sizes $(0, 3, 4)$, seven of sizes $(1, 3, 3)$, and ten of sizes $(2, 2, 3)$. These require one empty set, seven singletons, twenty pairs, twenty-four triples, and one four-set. All demands fit the binomial capacities, so $M_7 = 18$.

For eight coordinates, the construction uses one triple of sizes $(0, 4, 4)$, eight of sizes $(1, 3, 4)$, two of sizes $(2, 2, 4)$, and twenty-four of sizes $(2, 3, 3)$. Its 35 triples meet the weighted upper bound. Consequently a windmill with 19 outer triangles, and hence 58 vertices, requires exactly eight autonomous coordinates.

The companion program `verify_windmill_exact.py` performs two separate checks. First it verifies all prescribed rank demands, and the matching rational upper-bound certificates, for $2 \le d \le 300$. Second it constructs actual packings attaining M_d for $2 \le d \le 10$, converts them to graph codes, and checks distinctness and all cross-neighbor counts using integer arithmetic.

The construction implements the balancing operation underlying Bryant's theorem: equal-sized subsets exchange color classes according to an Euler orientation of an auxiliary multigraph. The code checks that the sum of squared coordinate degrees strictly decreases at each step. The resulting partitions, translated vertex codes, and arithmetic certificates are saved in

`windmill_exact_results.json`. For $d = 10$ this yields a code on W_{116}, which has 349 vertices. The $d = 3$ witness attains the capacity one but is not a minimum-length code for W_1, whose dimension is two.

These computations are finite checks of the formulas and explicit witnesses. The proof for all d is the upper-bound argument and the three capacity-verified constructions above.

References

[1] *Autonomy Dimension of Graphs: Spectral Separation, Product Activation, and Entropy Bounds*, research note, October 7, 2026, Section 7. Source at repository commit dae6532.

[2] D. Bryant, *On Almost-Regular Edge Colourings of Hypergraphs*, The Electronic Journal of Combinatorics **23**(4) (2016), Paper 4.7. doi:10.37236/5826.